PDF → CAD / DRAWING WORKBENCH

PDF to CAD User Guide

Convert complete drawings, or extract, calibrate and collect part contours

For the online workbench and local part-contour workspace · 2026-10-15

Two-point calibration example

1. Select outline endpoints A and B.
2. Enter the matching drawing length in mm.
3. Apply; check other critical dimensions.

Illustration only; no real product dimensions.

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The guide opens separately. Download results before refreshing, closing, or changing the workbench language.

01 · Choose your workflow

Drawing Workshop reads PDFs in your browser and creates editable DWG and DXF files. Convert complete pages, or extract part contours from different pages and arrange them in one CAD or PDF file.

Drawings are processed in the current page's memory. The user guide opens in a new tab so the original workbench can retain your files and calibrations.

Goal	Feature	Result
Keep full-page geometry and decodable text	Start conversion	One DWG / DXF pair per page
Trace the boundaries across a complete page	Trace whole page	One DWG / DXF pair per page
Keep only a lamp cup or other part's outer shape	Part contour collection	Several contours in one DWG, DXF or PDF

For isolated lamp-cup profiles, use Part contour collection: select a region → extract → calibrate two points → export the collection.

02 · Load drawings and choose a mode

Wait for "Conversion engine ready", then choose or drop files. Accepted inputs are PDF and ZIP / RAR archives containing PDFs. Other archive entries are ignored. Save scanned photos as PDF first.

Select a file to preview it, then choose the required file and page in the selectors. Remove password protection from a PDF before loading it.

Mode	When to use it
Automatic	Start here. The app chooses vector extraction or scan tracing for each page.
Vector PDF	For PDFs exported from CAD and similar software; preserves existing paths and decodable text where possible.
Scanned PDF	For image-based drawings; turns image boundaries into editable contours.

Scan defaults are 300 DPI and threshold 160. Increase the threshold to include lighter lines. Reduce DPI if processing is slow or memory runs short. Convert or extract again after changing these settings. A higher threshold can also include more noise.

03 · Calibrate each complete page

PDF paper dimensions usually differ from actual part dimensions. Use “Calibrate against a dimension” before full-page conversion. Each page has its own calibration.

Calibration scales the whole page uniformly. A page containing views at different scales cannot be fully calibrated from one reference. For individual outer shapes, extract and calibrate each part separately.

1. Choose a file and page in Drawing preview and show the source image.

2. Find a clear stated dimension. Click its two extension-line endpoints, carefully matching the intended measurement.

3. Enter the stated length in millimetres and select “Apply to this page”.

4. Check the reference length and page scale. Repeat for each page, then select “Start conversion”.

Calibration is required by default. If paper scale is sufficient, explicitly allow uncalibrated pages at paper / manual scale. These results are marked as dimensions unverified. A manually entered scale does not replace two-point calibration.

04 · Select and extract one outer contour

After loading files, open the green Part contour collection workspace. Contours from different files and pages can be arranged in the same export.

This feature extracts one closed outer boundary of the main shape inside the selection. Internal lines, holes and section details are excluded. Use it to compare and organize outer profiles. Extraction rasterizes the source page and represents curved boundaries as line segments; native arcs are not preserved.

1. Select a file and page, then locate the required front view or other part view.

2. Drag a rectangle around one complete part, leaving a small blank margin. Avoid dimensions, text, drawing borders and other views.

3. Select “Extract selected outer contour”. Inspect the contour preview and compare recesses, steps and the overall shape with the source.

4. If the result includes dimension lines, misses an edge or selects the wrong object, select a new region and extract again. Remove unwanted older contours.

Faint edges, gaps and connected annotation lines can affect extraction. A successful extraction does not prove that the shape is correct. Inspect the preview before calibrating. “Edit contour” selects a part for naming, calibration or rotation; edit individual geometry later in CAD software.

05 · Calibrate the extracted part

Part calibration and page calibration are separate. Every extracted part needs its own two-point calibration, even if its source page was already calibrated.

Choose a clear, longer stated dimension, such as maximum diameter or total height. The 60 mm illustration is a synthetic example, not a real product specification. A model name is not a dimension; enter the actual stated measurement.

1. On the extracted contour, click endpoints A and B corresponding to a known dimension. Pointer and touch clicks snap to nearby contour edges.

2. Enter its stated length, for example 60 mm, and select “Calibrate this part”.

3. Check the stated and measured reference lengths and the displayed width and height. Compare another dimension with the source.

4. Enter a model or name. Use “Rotate 90°” for sideways views; rotation preserves the calibrated reference length.

5. To correct the measurement, select “Reset calibration” and choose the points again. An unfinished new calibration cannot be exported.

Two-point calibration verifies only the selected reference distance and scales the complete shape uniformly. It does not automatically correct scan stretch, perspective distortion or other dimensional errors.

06 · Export several contours together

Repeat selection, extraction and calibration to build your collection. Check the parts to include, uncheck those to omit, or remove unwanted entries.

All selected parts must be calibrated before the DWG / DXF / PDF export buttons become available. Export one format at a time. Files go to your browser’s configured download location.

Format	Layout and scale	Use
DWG	One AutoCAD 2013 file; millimetre units and calibrated actual-size geometry at 1:1.	Edit and measure in compatible CAD software.
DXF	One editable exchange file; millimetre units and calibrated actual-size geometry at 1:1.	Import into CAD software for further work.
PDF	Landscape A4 with up to six parts per page and automatic page breaks; one shared display scale across the whole collection.	View, compare and print contour collections.

PDF contours are scaled to fit the sheet, with display scale and bounding dimensions shown. Check the scale when printing or measuring. Use the reference dimensions shown in the page, the original drawing and CAD measurements to verify actual dimensions.

07 · Download full-page results

Start conversion processes every page of every loaded file, generating DWG and DXF per page. The preview page does not limit this scope. Lines and curves can be selected separately. Decodable vector text is preserved where possible; scanned text remains an outline. Dimension labels are not restored as associative CAD dimensions.

Trace whole page also processes every page of every loaded file, using the current mode and page calibration. It omits native vector text, but text already converted to paths, scanned text and dimension numbers may remain. Thick scanned lines may produce outlines on both sides.

1. Check the success and failure counts. Review pages with warnings individually.

2. Select DWG or DXF on a page’s result to download that file.

3. Choose “Download all DWG”, “Download all DXF” or “Download both formats” to download the corresponding ZIP archive.

4. Extract the archive and inspect the files in CAD software. Check scale, lines, text and reference dimensions. The result preview is the source reference, not a rendered view of the exported CAD.

Bulk archives contain successfully generated results only. Bulk full-page download packages separate files. To place multiple outer shapes in one CAD drawing, use Part contour collection.

08 · Save your work and understand the limits

Drawings are not uploaded to a conversion server. No conversion history is stored or displayed. Files, previews, calibrations and temporary results exist only in the current workbench tab. Refreshing, closing, changing the workbench language, clearing the session or using Cancel and clear removes them; reload source files to start again.

Downloaded files remain on your device; clearing the workbench does not delete them. Download everything you need before clearing or closing the page. This guide can be read independently without loading the conversion engine.

Item	Current limit
Input	60 MiB per PDF; 50 PDFs, 100 pages and 200 MiB total per batch.
Scanning	30 million rendered pixels per page; 250,000 independent entities per page by default.
Part collection	Up to 100 contours per session and 4 million pixels per crop. Reduce the selected region or DPI if extraction is too complex.
Browser	A modern browser with WebAssembly, Web Worker and OffscreenCanvas support, accessed through a web address.

The first visit downloads program resources. Complex drawings require more memory. For large or slow jobs, process fewer files, reduce DPI or split the PDF first.

09 · Troubleshooting

Issue	What to do
Engine never becomes ready	Check your connection and browser support. Save any available results before refreshing; a refresh clears the session.
File fails to load or archive is empty	Check that it is a valid PDF, or a ZIP / RAR containing PDFs. Remove PDF password protection and check size and page limits.
Vector mode fails on a scan	Choose Automatic or Scanned PDF and convert again.
Contour contains annotations or the wrong shape	Select one complete part, avoid connected annotations and inspect gaps in the source. Adjust the threshold and extract again.
Contour export buttons are disabled	Select at least one part and complete two-point calibration for every selected part.
Downloaded dimensions still differ	Check endpoints, millimetre units and reference length. Verify a second dimension. Correct remaining scan distortion in CAD.
Download cannot be found	Check the browser’s download list, destination and download permission. Keep the workbench open while retrieving temporary results.
DWG opens incorrectly	Try the DXF from the same result. Check support for AutoCAD 2013 DWG and verify geometry and text.

If a page fails, the remaining successful results can still be downloaded.

10 · Check before delivery

Use this sequence to catch missing pages, incorrect scale and selection mistakes. The software extracts and organizes geometry; verify final engineering dimensions against the source drawing.

1. Confirm the file, page, complete part boundary and model label.
2. Confirm every selected part is calibrated and its reference matches the original dimension.
3. Check another diameter or height, plus recesses, steps and edge details.
4. Confirm millimetre CAD units, individually editable lines and an appropriate PDF display scale.
5. Confirm all required files are saved locally before clearing or closing the workbench.

Guide version: 2026-10-15. This guide describes the current online release. All diagram and dimension examples are synthetic.